

Algebra Basics

AI Tutor - Mathematics - Lesson 2

Understanding Variables

In algebra, letters like x and y stand in for numbers we don't know yet. For example, in the equation $x + 5 = 12$, the variable x represents the unknown number that makes the equation true (in this case, $x = 7$).

Solving Simple Equations

To solve an equation, isolate the variable on one side by performing the same operation on both sides. This keeps the equation balanced.

- $x + 5 = 12 \rightarrow$ subtract 5 from both sides $\rightarrow x = 7$
- $3x = 21 \rightarrow$ divide both sides by 3 $\rightarrow x = 7$
- $x / 2 = 4 \rightarrow$ multiply both sides by 2 $\rightarrow x = 8$

Order of Operations (PEMDAS)

- **P**arentheses first
- **E**xponents next
- **M**ultiplication and **D**ivision (left to right)
- **A**ddition and **S**ubtraction (left to right)

Practice Problem

Solve for x : $2x + 3 = 11$. Subtract 3 from both sides to get $2x = 8$, then divide both sides by 2 to find $x = 4$.